



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : CHE F214
Course Title : Engineering Chemistry
Instructor-in-charge : Prof. Krishna Etika
Tutorial Instructor : Prof. Krishna Etika and Prof. Shashank Muddu
Lecture Hour : T, Th, F 3rd Room 1203
Tutorial Hour : W 9th

1. Course Description:

Introduction to basic principles of various branches of chemistry and their applications in chemical Industries. Organic chemistry (Important functional groups and their reactions, Some important name reactions commonly used in the chemical industry); Physical chemistry (Thermodynamic processes, Thermochemistry, Electrochemistry, Adsorption, Catalysis), Chemical methods of analysis (volumetric analysis: neutralization, redox, complexometric titrations), Instrumental methods of analysis (IR, UV-vis spectroscopy, Chromatography), Engineering materials (cementing materials, refractories, polymers), Metals and alloys (Iron and steel), Corrosion (types of corrosion and corrosion control methods), Water treatment (Hardness of water, water analysis and water treatment), Fuels and combustion (Classification of fuels, Analysis of coal, Combustion, Calorific value and its determination).

2. Scope & Objective: The course aims to introduce various aspects of chemistry as applied to engineering. The course also aims to bridge the theoretical concepts and their practical engineering applications, thus highlighting the role of chemistry in the field of chemical engineering.

3. Text Books:

TB: Dr. Suba Ramesh and others, Engineering Chemistry, Wiley India, 2011,1st Ed.

4. Reference Books:

R1: P. W. Atkins, Elements of physical chemistry, 8th edition, Oxford University Press

R2: T. W. Graham Solomons and Craig B. Fryhle, Organic Chemistry, 9th edition, John Wiley and sons

R3: Perry and Green, Perry's Chemical Engineers' Handbook, 9th Edition, Section 2, McGraw Hill

R4: Dr S. S. Dara and Dr S. S. Umare, A Text book of Engineering Chemistry, S. Chand& Company

Ltd,2000 1st Ed. R5: William Kemp , Organic Spectroscopy, Palgrave, 1991,3rd Ed.



5. Course Learning Outcomes: Upon successful completion of this course, the student will be able to:

- Identify and illustrate key functional groups and industrially important organic reactions used in chemical synthesis.
- Explain the principles of thermodynamics and apply them to evaluate energy changes in chemical processes.
- Analyze electrochemical systems and interpret electrode potentials, electrochemical cell behavior, and battery performance.
- Describe adsorption processes and interpret isotherms and breakthrough curves relevant to industrial separations.
- Perform and evaluate chemical and instrumental analytical techniques such as titrations, spectroscopy (IR, UV-Vis), and chromatography.
- Classify polymers and engineering materials and assess their structure-property relationships for industrial use.
- Distinguish between various metals and alloys, especially steel, and justify their use based on physical and chemical properties.
- Assess the types and characteristics of fuels and calculate their calorific values using standard methods.
- Identify different types of corrosion and propose suitable prevention and control methods.
- Analyze water samples for impurities, determine hardness, and recommend appropriate water treatment techniques.
- Integrate chemical concepts to solve practical problems in industrial and environmental engineering contexts.
- Demonstrate scientific thinking, analytical reasoning, and technical communication skills through tests, quizzes, and project work.

6. Method of Instruction: Classroom lectures. It is recommended that the students attend the classes to understand the concepts. Tutorial sessions will focus only on specific topics and the topics covered in lectures will not be re-taught in the tutorial sessions.

7. Tentative Lecture Schedule:

Module Name	L. No	Topics	Learning Outcomes	Text /Ref Book Chapter #
Organic Chemistry	1-6	• Important Functional groups and their reactions: Alcohols, carboxylic acids, amines, aldehydes and ketones, ethers. • A Few Named Reactions: Friedel-Crafts acylation, Beckmann rearrangement, Aldol condensation, Cannizzaro's reaction, Hofmann rearrangement, Pinacol-Pinacolone rearrangement, Diels-Alder reaction.	• Identify and classify important functional groups in organic molecules (e.g., alcohols, amines, aldehydes). • Explain the mechanism of key industrially relevant named reactions (e.g., Friedel-Crafts acylation, Aldol condensation). • Predict the products of named organic reactions and illustrate their synthetic utility in industrial applications.	TB: Ch 9 R2



Physical Chemistry	7-9	Thermodynamics: Thermodynamic processes, Thermodynamic properties, First law of thermodynamics, Enthalpy, Heat capacity, Thermochemistry, Second law of thermodynamics, Entropy.	- Explain the laws of thermodynamics and apply them to chemical systems involving enthalpy, entropy, and heat capacity. - Differentiate between types of catalysis and analyze enzyme catalytic mechanisms.	TB: Ch 4 R3: sec 2
	10-11	Catalysis: Homogeneous and heterogeneous catalysis, autocatalysis, Enzyme catalysis, mechanism, catalytic promoters and inhibitors.	Construct electrochemical cells and calculate EMF using the Nernst equation.	TB: Ch 5
	12-16	Electrochemistry: Types of electrolytes, Electrochemical cells, Electrode potential, Galvanic cells, Nerst equation, Measurement of EMF, types of electrodes, concentration cells, Batteries.	Evaluate the behavior of various electrodes and classify types of concentration and galvanic cells.	TB: Ch 7
	17-19	Adsorption: Introduction to adsorption process, Adsorption isotherms, Breakthrough concentration curves.	Interpret adsorption isotherms and analyze adsorption data in context of separation processes.	TB: Ch 8
Analytical Chemistry	20-22	Chemical Methods of analysis: Volumetric analysis, Neutralization titrations, Redox titrations, Complexometric titrations.	- Perform volumetric titrations (acid-base, redox, and complexometric) and analyze titration curves. - Explain the principles of instrumental techniques such as IR, UV-Vis spectroscopy, and chromatography.	TB: Ch 11
	23-27	Instrumental Methods of analysis: Infrared spectroscopy UV-Visible spectroscopy Chromatography.	Compare the strengths and limitations of different analytical methods and recommend suitable techniques for specific applications.	TB: Ch 12 R5: Ch 4
Industrial & Engineering Chemistry	28-32	Fuels and Fuel analysis: Classification of fuels, Calorific value, combustion, solid fuels-coal, Proximate and ultimate Analysis of coal, liquid fuel, gaseous fuels, flue gas analysis.	- Classify solid, liquid, and gaseous fuels and explain their combustion characteristics. - Analyze coal using proximate and ultimate analysis and compute calorific values using standard techniques.	TB: Ch 16
	32-36	Water and its impurities: Impurities in water, Hardness of water and its determination, Analysis of water, Boiler feed water, Techniques for water softening.	- Identify common impurities in water and measure hardness using standard methods. - Evaluate industrial water quality and recommend suitable treatment and softening techniques for boiler feed water and wastewater.	TB: Ch 19 R4: Ch 1
	36-40	Polymers: Classification of Polymers, Types of polymerization, Molecular weight of polymers, polymerization techniques, some important commercial thermoplastics and thermosetting resins,	- Classify polymers based on structure, properties, and synthesis techniques. - Distinguish between thermoplastics and thermosets and evaluate their suitability for engineering applications.	TB: Ch 13
	40-42	Corrosion : Types of corrosion, Factors influencing rate of corrosion, Corrosion control methods, Protective coatings,	- Explain different types of corrosion and analyze the factors influencing corrosion rates. - Propose suitable corrosion control methods including coatings and inhibitors.	TB: Ch 18



6. Evaluation Scheme:

Component	Duration (minutes)	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Class Tests (3 out of 4) (Dates will be announced at least five days in advance)	40	30		CB and/or OB
Mid Examination	90	25		CB
AI Based Term Project	-	10		OB
Comprehensive Examination	180	35		CB

OB- Open Book, CB- Closed Book

- Chamber Consultation Hours: T 10th** (If not feasible, then email the I/C for an alternate time slot)
- NC Criteria:** **The minimum cut-off mark is at least 30% of the median marks of the class to obtain a valid grade. The students scoring below this threshold will be awarded NC.**
- Attendance Criteria:** It is expected that each student must attend all lectures and tutorials and participate in all the evaluative components.
- Notices:** The notices will be sent to the BITS email ID of the registered students through Google Classroom/Nalanda.
- Make-up Policy:** The granting of a makeup exam is a privilege and not a right! Make-up will be granted for genuine medical emergencies after thorough verification of provided documents. Merely submitting a prescription by a doctor recommending bed rest is not sufficient without diagnostic tests corroborating the doctor's advice. The doctor must issue a certificate on their official letter head stating the reasons for prescribing bed rest backed by relevant diagnostic confirmations. In case of out of campus medical consultation, the CMO of BITS Medical Center must certify the authenticity of your document and issue you a certificate on their official letter head. Except for emergency situations, prior permission of IC is compulsory. No makeup will be granted for evaluative components with a buffer.
- AI Based Term Project:** The students will be divided into a group of 3-4 students per team. Each team will be required to develop an AI based project on engineering chemistry applications. A few project titles are listed here but a team will be free to propose any project relevant to the course. The proposed project must be approved by the I/C prior to the initiation.
 - Predicting Water Quality Parameters Using Machine Learning
 - Spectral Data Classification Using AI
 - AI-Based Fuel Quality Estimation
 - Predicting Corrosion Risk Based on Environmental Factors
 - Polymer Property Prediction Using Molecular Descriptors
 - Reaction Condition Optimization Using AI



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- Automated Detection of Anomalies in Water Sample Data
 - AI-Based Prediction of Adsorption Efficiency
 - Chatbot Tutor for Engineering Chemistry Concepts
13. Lecture slides and other resources (if any) will be provided to only those who attend the lecture sessions. The lecture notes will not be uploaded on Nalanda.
14. *The lecture slides (if any) provided to the students registered in this course are for academic purposes only and are considered the intellectual property of BITS Pilani. Such materials should not be uploaded to any external websites/forums without the permission of the Instructor.*

Prof. Krishna C. Etika
Instructor-in-charge | **CHE F214**